

OKELLO ELLY

Software Developer | Systems Architecture | AI & Full-Stack Engineering

Scalable Systems • Full-Stack Engineering • Backend Development • Android Development • Cybersecurity • AI-Assisted Development

CONTACT INFORMATION

Location: Nairobi, Kenya

Email: iamellyokello@gmail.com

Phone: +254 769 182 575

LinkedIn: <https://www.linkedin.com/in/okelloelly/>

Portfolio: <https://okelloelly.vercel.app/>

GitHub: <https://github.com/Elly739>

PROFILE

Software Developer and Systems Architect with hands-on experience designing and building secure, scalable software systems across digital learning, campus services, civic technology, analytics, and operational platforms. Skilled in full-stack and backend development, REST APIs, database architecture, mobile application development, networking, cybersecurity, and secure software engineering, with a strong foundation in object-oriented programming, algorithm design, software testing, system design, and DevOps practices.

Founder and developer of VumaGrid, SmartWaste, and Pioneer Africa, applying software architecture, data-driven engineering, and systems thinking to real-world challenges. AI-fluent developer with experience in AI-assisted development, Claude API integration, AI workflow automation, and AI-powered product development using technologies including Claude, Base44, Lovable, LangGraph, CrewAI, Zapier, and n8n.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C#, C++, C, SQL, HTML, Kotlin

Frontend Development: HTML, CSS, JavaScript, React, XML

Backend Development: Python Flask, REST APIs, Backend Services, API Integration, JWT Authentication, RBAC

Mobile Development: React Native, Expo Go, Kotlin, Java, Android Studio, XML Layouts, Mobile UI Design, Application Architecture

Databases: MySQL, SQLite, PL/SQL, Relational Database Design

Networking & Security: TCP/IP, VLANs, Routing, Switching, IP Addressing, Network Configuration, Linux Fundamentals, Cybersecurity Fundamentals, Secure System Design

Software Engineering: Object-Oriented Programming, Data Structures and Algorithms, Software Testing, Software Metrics, Quality Assurance, System Analysis, System Design and Architecture

Enterprise & Desktop Development: C#, ASP.NET, Windows Forms, Console Applications

Tools: Git, GitHub, AI-Assisted Development Tools

PROFESSIONAL EXPERIENCE

AI & Mobile Development Intern — Momentum Labs

August 2026 – Present

- Contributing to **AfroNative**, an AI-powered African language learning platform built with **React Native** focused on practical, mobile-first language education.
- Developing mobile-first learning experiences incorporating native African languages, pronunciation feedback, conversational learning, and interactive lessons.

Front-End AI Engineering Intern — FlyRank AI

July 2026 – August 2026

- Contributed to frontend engineering initiatives within AI-driven software development projects.
 - Developed and refined user interfaces using modern frontend technologies and component-based development practices.
 - Collaborated on feature implementation, testing, documentation, and product improvements.
-

EDUCATION

Bachelor of Science in Computer Software Engineering

Zetech University

January 2025 – November 2027

MAJOR ENGINEERING PROJECTS

VumaGrid — Founder & Systems Architect

AI-Powered Distributed Energy Management System (DERMS)

2026 – Present

- Founded and architecting an AI-powered Distributed Energy Management System (DERMS) designed to optimize energy usage in commercial buildings.
 - Designing system architecture for monitoring, analytics, and intelligent decision-making across distributed energy resources.
-

SmartWaste — Founder & Software Developer

Smart Waste Management Platform

2026 – Present

- Founded and developed a full-stack digital waste management platform focused on improving waste collection visibility, tracking, and operational reporting.
- Designed backend systems and APIs supporting waste data collection, management, analytics, and reporting workflows.

BIT System — Backend Developer & System Architect

2024 – Present

Technologies: Python (Flask), MySQL, SQLite, JWT, REST APIs, HTML, CSS

- Designed and developed a campus digital platform supporting student services, event management, payments, analytics, and administrative workflows.

Pioneer Africa — Founder & Technical Lead

Africa Rise Ecosystem

2026 – Present

- Founded a digital ecosystem focused on bridging gaps between learning, skills development, mentorship, and real-world opportunities for African students and young innovators.
- Designed platform architecture supporting learning pathways, mentorship connections, innovation programs, and opportunity discovery.

ADDITIONAL ENGINEERING PROJECTS

LearnLight — Software Developer

2026

- Contributed to the development of a digital learning platform supporting structured online education and accessible learning experiences.

Campus Pulse — Software Developer

2026

Technologies: HTML, CSS, JavaScript, Kotlin, XML

- Developed a university engagement platform connecting students with campus events, hackathons, sports tournaments, career fairs, and community activities.

Nairobi Hidden Gems — Software Developer

2026– Present

Technologies: HTML, CSS, JavaScript, Kotlin, XML

- Developed a social discovery platform enabling users to explore aesthetic locations and experiences across Nairobi.

CERTIFICATIONS

Introduction to Modern AI — Cisco Networking Academy (2026)

- Artificial intelligence fundamentals
- Machine learning concepts

Junior Cybersecurity Analyst — Cisco Networking Academy

- Threat analysis
- Vulnerability assessment

Huawei Computer Networks (CRA Training Program)

- Network fundamentals

Cisco Certified Cybersecurity Essentials

- Cyber risk management

Data Science and Analytics — HP LIFE (2025)

- Data visualization

AI Fluency: Framework and Foundations — Anthropic (2025)

- Prompt engineering
- AI workflows
- AI-assisted software development

AI Fluency for Builders — Anthropic Issued 2026

- AI delegation, AI-assisted development, responsible AI workflows, and developer productivity.

Teaching the AI Fluency Framework — Anthropic Issued 2026

- AI fluency frameworks, AI literacy, and practical AI adoption.

Claude 101 — Anthropic**Issued 2026**

- AI-assisted research, Claude workflows, and productivity applications.

AI Safari — Power Learn Project**Issued 2026**

- Artificial intelligence applications, prompt engineering, AI agents, workflow automation, responsible AI, design thinking, ChatGPT, Claude, Gemini, Zapier, n8n, CrewAI, and LangGraph.

Building with the Claude API — Anthropic**Issued 2026**

- Claude API integration
 - Building AI-powered applications
 - API-based generative AI development
-

VOLUNTEER LEADERSHIP & PROFESSIONAL DEVELOPMENT**Aspire Leaders Program — Graduate, Cohort 1 (2026)**

- Completed leadership development program focused on leadership, systems thinking, problem solving, and professional growth.

Aspire Leaders Program — Moderator, Cohort 2 and 3 (Volunteer) (2026)

- Assisted with program operations and peer learning activities.

Secretary — Innovation Club, Zetech University

- Supported innovation initiatives, technical activities, and student engagement programs.

GMI Mentorship Program (2026)

- Participated in mentorship focused on leadership development and career growth.

Anticipatory Governance: Systems Thinking and Strategic Foresight (Parts 1 & 2) — 2026

- Systems thinking frameworks and strategic foresight implementation.

McKinsey Forward Program — 2026

- Professional development focused on communication, leadership, and workplace effectiveness.

Cisco Networking Academy Learn-A-Thon Recognition — (2025& 2026)

- Recognized twice for participation and completion of Cisco learning activities supporting technology skills development